

Fundamentals of Computer

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The Computer Defined

- ▶ Electronic device
- ▶ Converts data into information
- ▶ Modern computers are digital
 - ▶ Two digits combine to make data
- ▶ Older computers were analog
 - ▶ A range of values made data



Computers For Individual Use

- ▶ **Desktop computers**

- ▶ The most common type of computer
- ▶ Sits on the desk or floor
- ▶ Performs a variety of tasks

- ▶ **Workstations**

- ▶ Specialized computers
- ▶ Optimized for science or graphics
- ▶ More powerful than a desktop



Computers For Individual Use

- ▶ **Notebook computers**
 - ▶ Small portable computers
 - ▶ Weighs between 3 and 8 pounds
 - ▶ About 8 ½ by 11 inches
 - ▶ Typically as powerful as a desktop
 - ▶ Can include a docking station



Computers For Individual Use

- ▶ **Tablet computers**
 - ▶ Newest development in portable computers
 - ▶ Input is through a pen
 - ▶ Run specialized versions of office products



Computers For Individual Use

- ▶ **Smart phones**
 - ▶ Hybrid of cell phone and Personal Digital Assistants
 - ▶ Very small computers
 - ▶ Can do anything same as tablet computer
 - ▶ Can use GPS, GSM Sim and other mobile phone services
 - ▶ Web surfing, e-mail access



Computers For Organizations

- ▶ Network servers
 - ▶ Centralized computer
 - ▶ All other computers connect
 - ▶ Provides access to network resources
 - ▶ Multiple servers are called server farms
 - ▶ Often simply a powerful desktop



Computers For Organizations

▶ Minicomputers

- ▶ Called midrange computers
- ▶ Power between mainframe and desktop
- ▶ Handle hundreds of users
- ▶ Used in smaller organizations
- ▶ Users access through a terminal



Computers For Organizations

▶ Mainframes

- ▶ Used in large organizations
- ▶ Handle thousands of users
- ▶ Users access through a terminal
- ▶ Extremely high throughput and reliability



Computers For Organizations

- ▶ **Supercomputers**
 - ▶ The most powerful computers made
 - ▶ Handle large and complex calculations
 - ▶ Process trillions of operations per second
 - ▶ Found in research organizations



Computers For Organizations

▶ Embedded Computer

- ▶ These are specialized microcontrollers built into other machines to perform a single, dedicated function
- ▶ Usage: Car navigation systems, smart thermostats, industrial robots, and medical devices.
- ▶ Example: The chip in your microwave or the flight control system in a drone.



Computers In Society

- ▶ **More impact than any other invention**
 - ▶ Changed work and leisure activities
 - ▶ Used by all demographic groups
- ▶ **Computers are important because:**
 - ▶ Provide information to users
 - ▶ Information is critical to our society
 - ▶ Managing information is difficult



Computers In Society

- ▶ **Computers at home**
 - ▶ Many homes have multiple computers
 - ▶ Most American homes have Internet
 - ▶ Computers are used for
 - ▶ Business
 - ▶ Entertainment
 - ▶ Communication
 - ▶ Education



Computers In Society

- ▶ **Computers in education**
 - ▶ Computer literacy required at all levels
- ▶ **Computers in small business**
 - ▶ Makes businesses more profitable
 - ▶ Allows owners to manage
- ▶ **Computers in industry**
 - ▶ Computers are used to design products
 - ▶ Assembly lines are automated



Computers In Society

- ▶ Computers in government
 - ▶ Necessary to track data for population
 - ▶ Police officers
 - ▶ Tax calculation and collection
 - ▶ Governments were the first computer users



Computers In Society

- ▶ **Computers in health care**
 - ▶ Revolutionized health care
 - ▶ New treatments possible
 - ▶ Scheduling of patients has improved
 - ▶ Delivery of medicine is safer



How two digits combination make data

- ▶ Computer store data as binary number (0,1)
- ▶ Everything we pressed in Keyboard convert to binary
- ▶ Use ASCII code to convert data to binary



Common Number Systems

System	Base	Symbols	Used by humans?	Used in computers?
Decimal	10	0, 1, ... 9	Yes	No
Binary	2	0, 1	No	Yes
Octal	8	0, 1, ... 7	No	No
Hexa-decimal	16	0, 1, ... 9, A, B, ... F	No	No
Penta	5	0,1,2,3,4	No	No



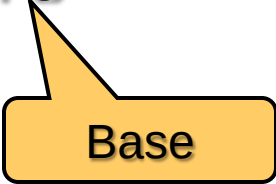
Quantities / Counting

Decimal	Binary	Octal	Hexa- decimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7



Quick Example

$$25_{10} = 11001_2 = 31_8 = 19_{16}$$



Base

How computer store data

- ▶ The data below has been written in an 8-bit computer.
- ▶ Retrieve what is written there.

01010000 01011001 01010100 01001000 0100111101001110



How computer store data?

01010000 01011001 01010100 01001000 0100111101001110

A	65	G	71	M	77	S	83	Y	89
B	66	H	72	N	78	T	84	Z	90
C	67	I	73	O	79	U	85		
D	68	J	74	P	80	V	86		
E	69	K	75	Q	81	W	87		
F	70	L	76	R	82	X	88		





End of Inside the Computer